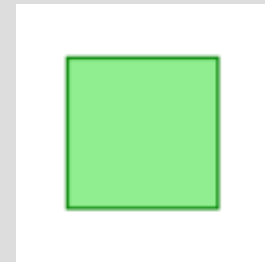
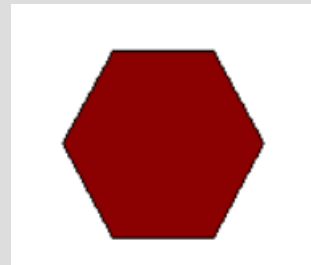
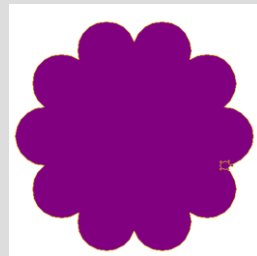


ים  כתיבת פעולות שמקבלות



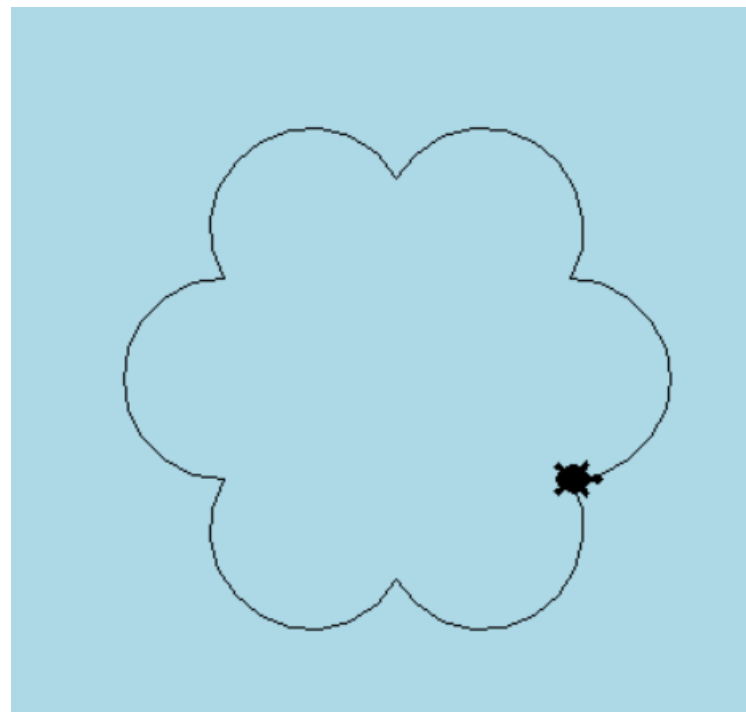
פעולה שיודעת לקבל דמות ולצייר
בעזרתה פרח

```
def draw_flower(player):  
    for i in range(6):  
        player.circle(50,180)  
        player.rt(120)
```

```
player = turtle.Turtle()  
player.shape('turtle')
```

```
draw_flower(player)
```

קריאה לפעולה



```

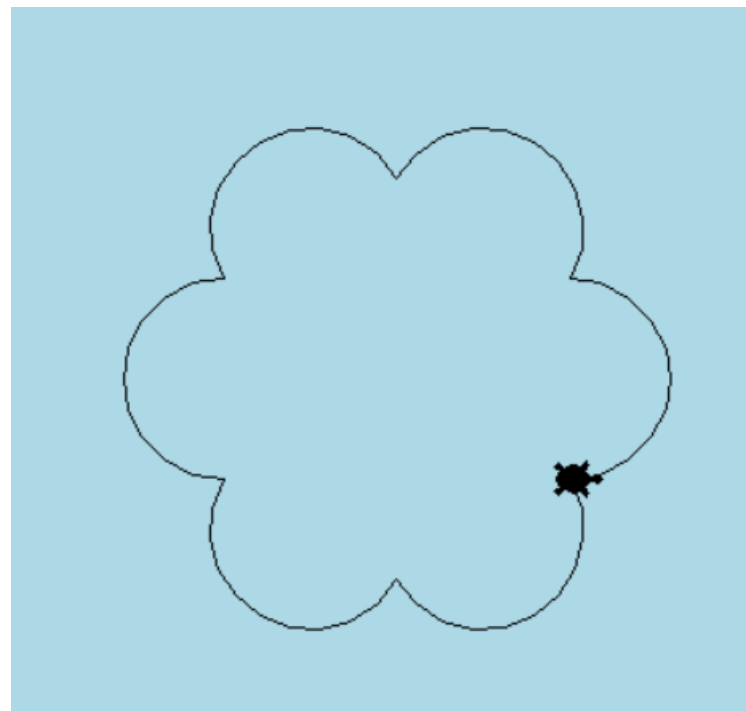
def draw_flower2(player):
    radius = random.randint(a: 10, b: 80)
    for i in range(6):
        player.circle(radius, 180)
        player.rt(120)

    player = turtle.Turtle()
    player.shape('turtle')

    draw_flower2(player)

```

נגריל את
גודל עלי
הכותרת
של
הפרח

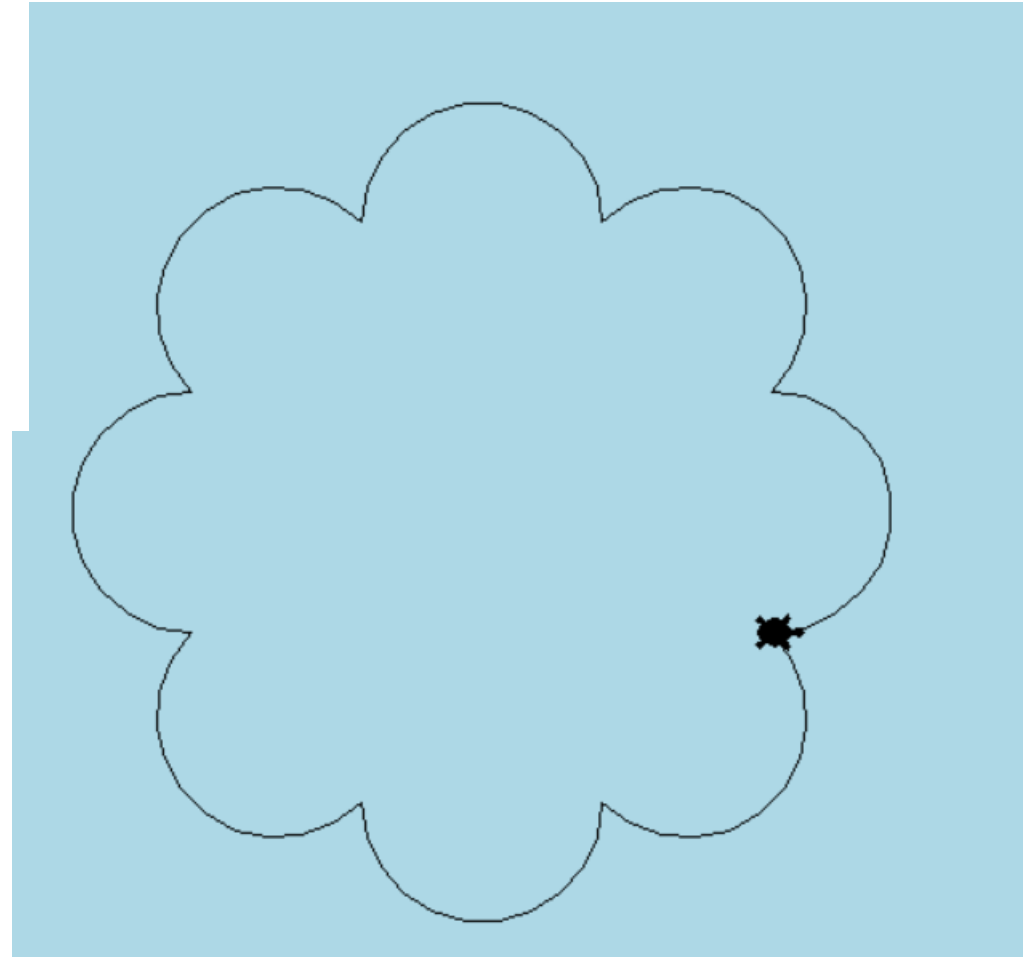


```
def draw_flower3(player):  
    radius = random.randint(a: 10, b: 80)  
    num_petals = random.randint(a: 3, b: 12)  
    angle = 180 * (num_petals - 2) / num_petals  
    for i in range(num_petals):  
        player.circle(radius, 180)  
        player.rt(angle)
```

נגריל את מספר עלי
הכותרת ונחשב את
הזווית ביניהם בהתאם

```
player = turtle.Turtle()  
player.shape('turtle')
```

```
draw_flower3(player)
```



```

def draw_flower4(player):
    radius = random.randint(a: 10, b: 80)
    num_petals = random.randint(a: 3, b: 12)
    angle = 180 * (num_petals - 2) / num_petals
    f_color = random.choice(['pink', 'lightblue', 'purple', 'yellow', 'lightgreen', 'turquoise'])
    player.color(f_color)
    player.fillcolor(f_color)
    player.begin_fill()
    for i in range(num_petals):
        player.circle(radius, 180)
        player.rt(angle)
    player.end_fill()

```

נגריל גם את צבע הפרח

```

player = turtle.Turtle()
player.shape('turtle')

draw_flower4(player)

```



```
def draw_flower5(player):
    radius = random.randint(a: 10, b: 80)
    num_petals = random.randint(a: 3, b: 12)
    angle = 180 * (num_petals - 2) / num_petals
    f_color = random.choice(['pink', 'lightblue', 'purple', 'yellow', 'lightgreen', 'turquoise'])
    player.color(f_color)
    player.fillcolor(f_color)
    player.begin_fill()
    x_cor = random.randint(-350, b: 350)
    y_cor = random.randint(-350, b: 350)
    player.pu()
    player.goto(x_cor, y_cor)
    player.pd()
    for i in range(num_petals):
        player.circle(radius, 180)
        player.rt(angle)
    player.end_fill()

player = turtle.Turtle()
player.shape('turtle')

draw_flower5(player)
```

הפרח לא חייב להיות
באמצע. ניתן להגדיל
את שיעור המיקום שלו
על המסך



```
def random_color(player):
    f_color = random.choice(['pink', 'lightblue', 'purple', 'yellow', 'lightgreen', 'turquoise'])
    player.color(f_color)
    player.fillcolor(f_color)
    player.begin_fill()
```

2 usages

```
def random_goto(player):
    x_cor = random.randint(-350, 350)
    y_cor = random.randint(-350, 350)
    player.pu()
    player.goto(x_cor, y_cor)
    player.pd()
```

```
def draw_flower6(player):
    radius = random.randint(10, 80)
    num_petals = random.randint(3, 12)
    angle = 180 * (num_petals - 2) / num_petals
    random_color(player)
    random_goto(player)
    for i in range(num_petals):
        player.circle(radius, 180)
        player.rt(angle)
    player.end_fill()
```

מכיוון שהמיקום והצבע
פועלים על הדמות, ניתן
להגריל אותם בפעולות
נפרדות.
אולי בהמשך אף נצטרך
פעולות אלא בשביל
ציורים אחרים

```
player = turtle.Turtle()
player.shape('turtle')
```

```
draw_flower5(player)
```



```
player = turtle.Turtle()
player.shape('turtle')
player1 = turtle.Turtle()
player1.shape('turtle')
player2 = turtle.Turtle()
player2.shape('turtle')

draw_flower6(player)
draw_flower6(player1)
draw_flower6(player2)
```

כאן אנחנו מפעילים את
אותה הפעולה עבור
שלוש דמויות שונות,
ומקבלים תוצאות שונות
באותה תוכנית

